AI Forward Deployed Engineer

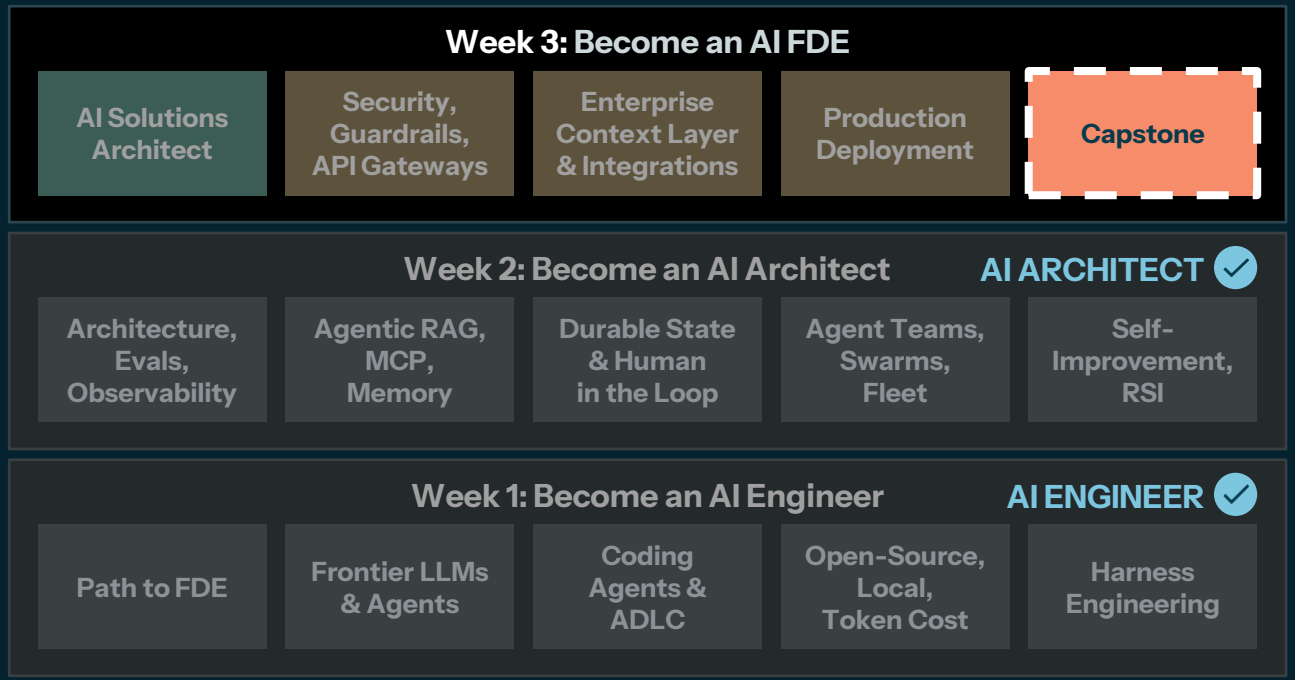
The AI Architect & FDE Course



Week 3 Day 5

3 weeks to Forward Deployed Engineer

Theory Practical Projects



Taking stock

You can already:

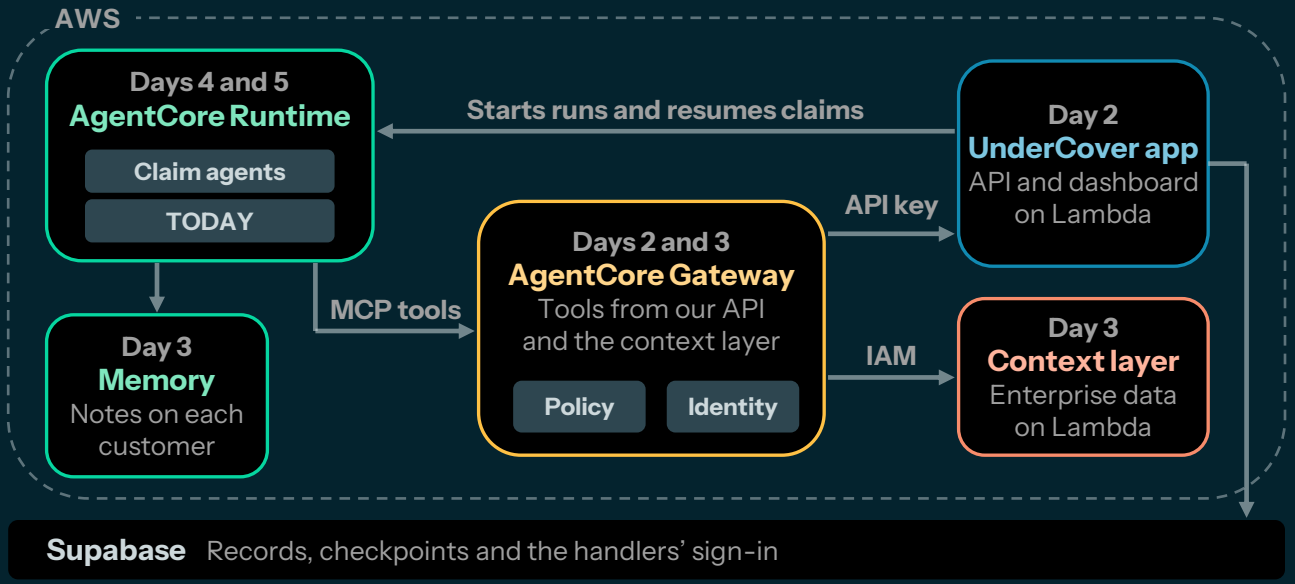
- ✓ Deliver solutions that measurably solve business problems with evals and observability
- ✓ Create durable, scalable agents with memory and a human in the loop
- ✓ Drive an AI implementation from discovery to delivery and maintenance
- ✓ Deploy an enterprise agent with a gateway, context layer & memory on a runtime

By the end of today, you'll be able to:

- ○ Confidently build agents with durable state, memory and runtimes
- ○ Hand over an agent solution with evidence against evals / success criteria
- ○ Lay out the next steps to deliver impact as an AI Architect and FDE
- ○ Say that you are an AI Engineer, an AI Architect and an AI FDE!

Recap: what we built

And the one last thing



Models still come from OpenRouter

The UnderCover Copilot

A copilot agent for claims handlers



Durable state

LangGraph checkpoints in Supabase

Pick up any conversation later

Asks the handler before paying



Production runtime

AgentCore Runtime

A session per conversation

Keeps working after the reply



Namespaced memory

AgentCore Memory

A namespace per handler

Remembers each handler's preferences

A LangGraph agent loop with the claim agent's tools and the handler's login

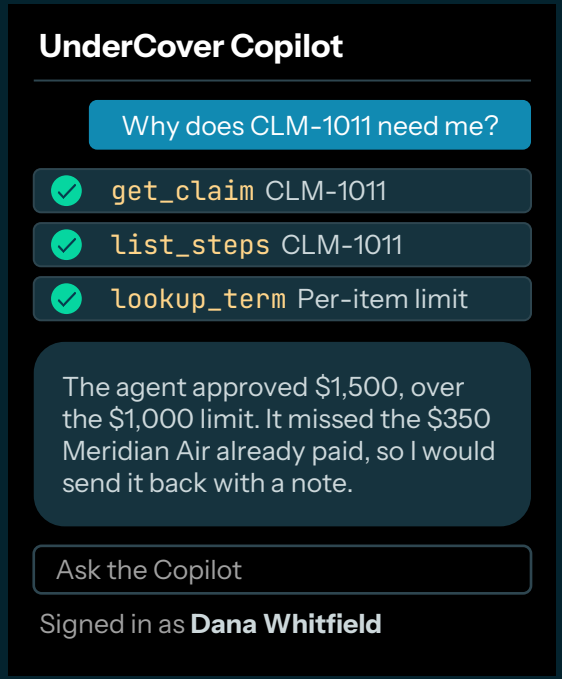
The Copilot sidebar

The handler sees each step as the Copilot works

Tool calls as cards
Each one appears as it happens

The answer in full
It appears when the turn is done

The handler's login
It can do only what the handler can do



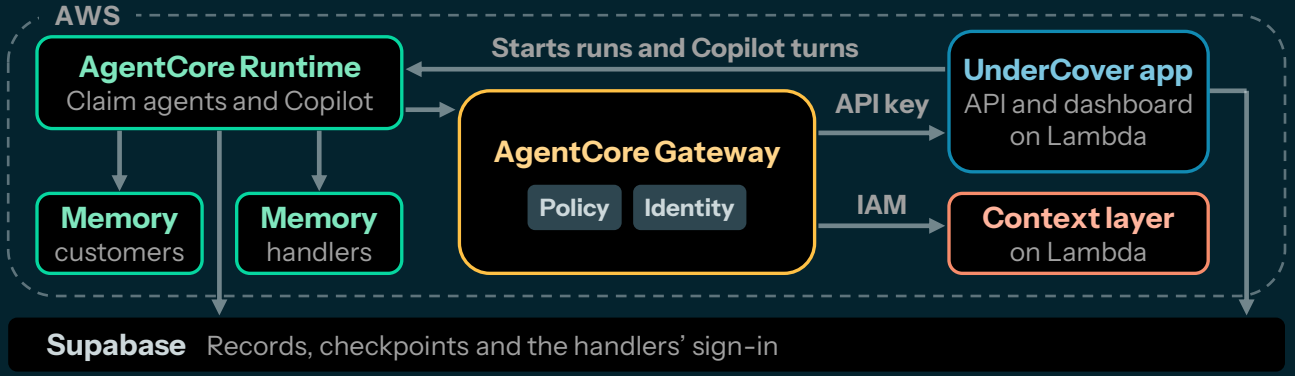
Building it with Claude Code

The change is written up in labs/week3day5_product_COPILOT.md

1. **Copy the brief into docs**
   `cp labs/week3day5_product_COPILOT.md docs/COPILOT.md`
2. **Ask Claude Code for a plan**
   Please plan the Copilot in docs/COPILOT.md
3. **Let it build and commit each phase**
   Please implement the plan for the Copilot
4. **Or check out my version**
   `git checkout week3day5-start` or `git checkout week3day5-end`

Lab 1: UnderCover in production

Deploying all of UnderCover to AWS



1. **Deploy the new app**
   The Copilot's routes and the sidebar
2. **Bring back Day 3's pieces**
   The Gateway, Policy and context layer
3. **Create both Memories**
   The claim agent's and the Copilot's
4. **Deploy both runtimes**
   `agentcore deploy` with their settings
5. **Point the app at both runtimes**
   Then reseed and call each one
6. **Talk to the Copilot**
   The handover conversation

The handover conversation

Five questions for the deployed Copilot

1. “Work the queue with 10 agents and tell me what needs me”
   A dispatch on Runtime while the dashboard updates
2. “Why does CLM-1011 need me?”
   Its steps, the policy wording and the payout limit
3. “Send it back with a note”
   The claim agent picks it up in a new session
4. “Sign off on another one”
   The Copilot asks before it pays
5. “Are we meeting the success criteria we agreed?”
   An evaluation on the held-out claims

The success criteria we agreed

Straight through
A share of claims handled with no handler

Accuracy
At least as accurate as the handlers

Over the limit
Every payout approved by a person

Your assignment

Build your own LangGraph solution on AgentCore Runtime

A business problem where agents make decisions within limits, like approving a claim



Decisions with limits

A person signs off above the limit



Durable state

LangGraph checkpoints in a database



Enterprise context layer

Definitions, rules and reference data for the agent



Long-term memory

AgentCore Memory with a namespace per user



Runtime deployment

Deployed on AgentCore Runtime

Most importantly: be the Forward Deployed Engineer -
start by defining evals, and end by measuring them to determine success

Lab 2: The final teardown

Deleting everything and checking what is left

What it deletes

AgentCore
Both runtimes, the Gateway and its Policy, the credential providers and both Memories

Lambda
The app and the context layer, with their roles and logs

Account setup
The CLI's encryption key and the CDK bootstrap

The three checks



UnderCover's resources
Each service checked by name



Everything on the account
AWS Resource Explorer, by service and Region



What it cost
Today and the last 30 days, from your budgets

Everything you can do

Week 1
AI Engineering

Agent loops and tools
Frontier LLMs and agents
Model selection
Agent frameworks
Coding agents
The ADLC
Open-source & local models
Token cost
Context engineering
Harness engineering

Technologies

OpenAI SDK
OpenRouter
LangChain
Claude Code
Claude Agent SDK

Week 2
AI Architecture

System design
Agent architecture
Evals and observability
Cost and speed trade-offs
Agentic RAG and memory
MCP
Durable state
Human in the loop
Agent swarms
Self-improvement

Technologies

Langfuse
Qdrant / vector db
MCP
LangGraph
Supabase

Week 3
Forward Deployed Engineering

Customer discovery
Governance & compliance
Security and guardrails
API and MCP gateways
Enterprise context layer
Enterprise integrations
Cloud infrastructure
Production deployment
Operational handover
Adoption

Technologies

Serverless
AWS AgentCore Gateway
AWS AgentCore Policy
AWS AgentCore Memory
AWS AgentCore Runtime

What you've learned

Week 1
AI Engineer

Identify the skills of an AI Architect and Forward Deployed Engineer (FDE) with the roadmap to get there

Accelerate delivery with Claude Code and the Agentic Development Lifecycle (ADLC) while staying accountable

Apply Harness Engineering to build agents with LangChain, Deep Agents and the Claude Agent SDK

Choose between frontier, open-source and local LLMs on intelligence, speed, token cost and privacy

Week 2
AI Architect

Use evals and observability in Langfuse to drive agent architecture decisions and measure business outcomes

Give agents long-term memory with Agentic RAG over MCP using vector datastores

Deliver production-grade agent workflows in LangGraph with durable state, crash recovery and human in the loop

Scale and optimize with agent swarms, LLM as a judge and recursive self-improvement

Week 3
AI FDE

Lead AI initiatives with your clients from discovery and metrics to governance, responsible AI and handover

Secure agents against prompt injection and the lethal trifecta with guardrails, policies and AI gateways

Create a data and integration architecture for agents with an enterprise context layer and MCP gateway

Deploy LangGraph agents to production with AWS Bedrock AgentCore Runtime, Gateway, Policy and Memory

The five areas in job adverts

Share of 316 job adverts that mention each area



88%

AI Engineering



79%

AI Architecture



95%

Forward Deployed Engineering



88%

Solved real-world business problems



78%

Delivered to customers

Where you stand on each area



Covered in Week 1



Covered in Week 2



Covered in Week 3



Next, with your assignment



Next, with a real user

Source: our review of 316 adverts from 184 employers for FDE, AI Architect and related roles

The next step

Build real-world projects to demonstrate your expertise

1

Start with your assignment

Your own solution, deployed

2

Solve a problem for a real user

Technical consulting or a project for an internal customer

3

Show the work

A public repo, a demo video and a write-up

The two areas of hands-on experience from Day 1

Solved real-world business problems

Delivered to customers

3 weeks to Forward Deployed Engineer

Theory Practical Projects

Week 3: Become an AI FDE

AI Solutions Architect	Security, Guardrails, API Gateways	Enterprise Context Layer & Integrations	Production Deployment	Capstone

Week 2: Become an AI Architect — AI ARCHITECT

Architecture, Evals, Observability	Agentic RAG, MCP, Memory	Durable State & Human in the Loop	Agent Teams, Swarms, Fleet	Self-Improvement, RSI

Week 1: Become an AI Engineer — AI ENGINEER

Path to FDE	Frontier LLMs & Agents	Coding Agents & ADLC	Open-Source, Local, Token Cost	Harness Engineering

3 weeks to Forward Deployed Engineer

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Theory Practical Projects



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AI ARCHITECT

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Week 1: Become an AI Engineer

AI ENGINEER

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Congratulations!

Remember: I come with the package!



- Post your certificate and your assignment on LinkedIn
- Tag me and I'll amplify your success
- Consider my other AI courses if you'd like to go deeper
- Most importantly: **solve real world problems for your clients**

AI Forward Deployed Engineer

The AI Architect & FDE Course



100% Complete